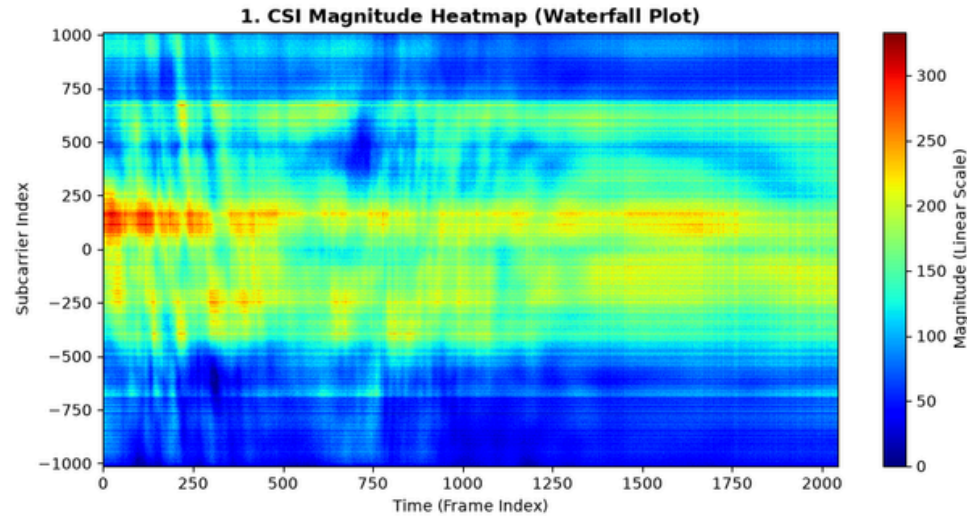
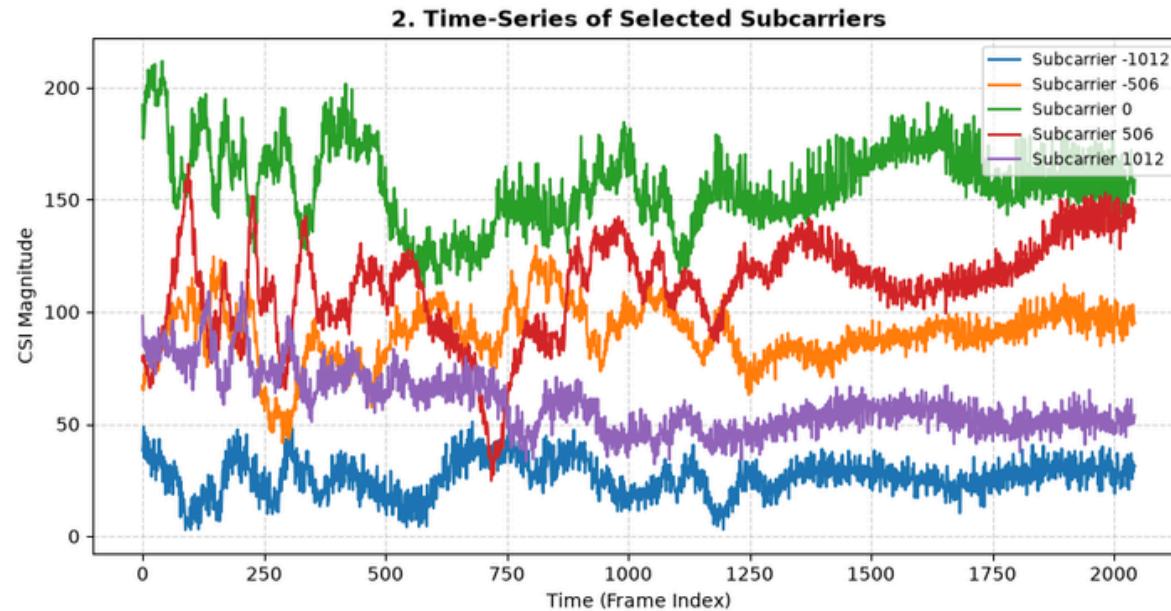




1. CSI Magnitude Heatmap (Waterfall Plot)

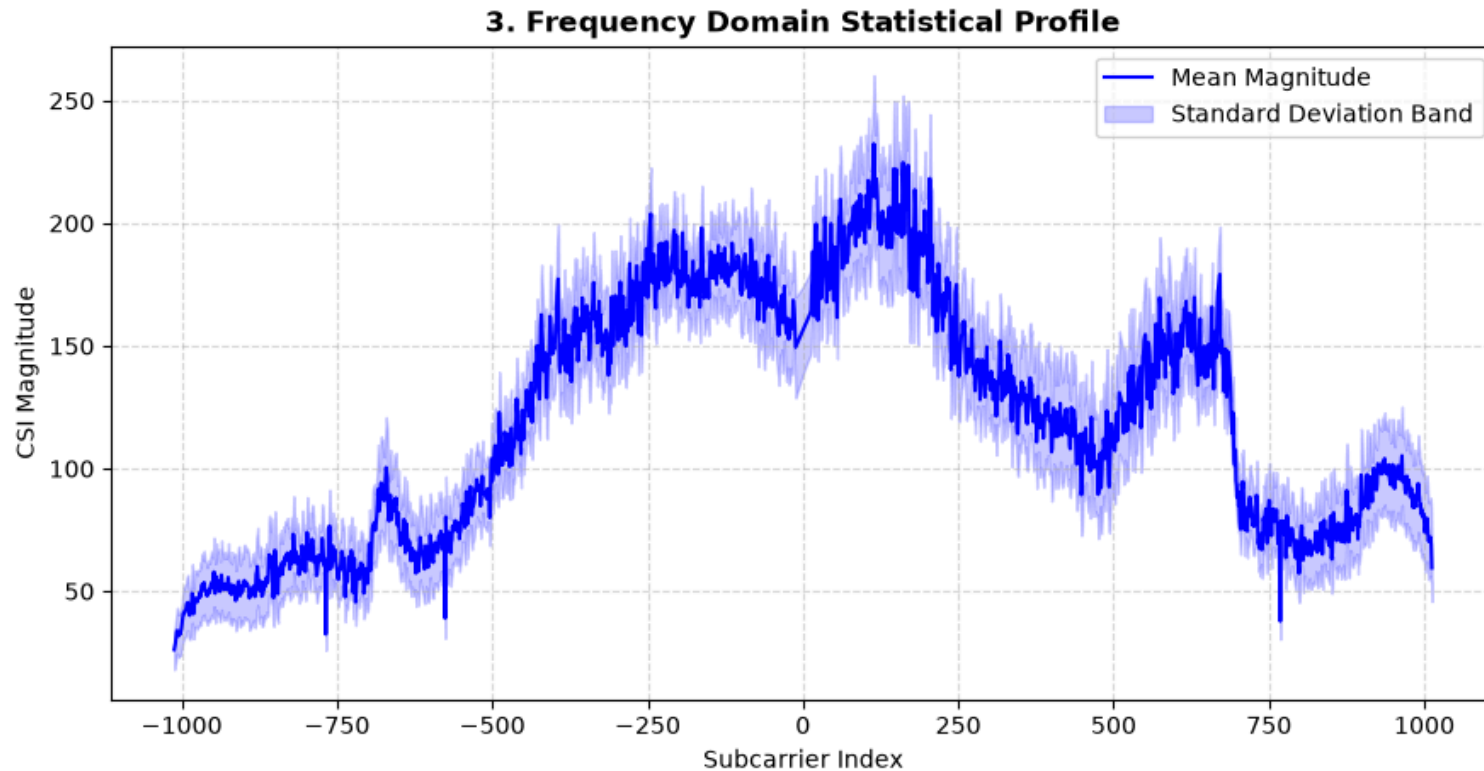
What it is: A 2D macro-view of your entire experiment. The X-axis represents time, the Y-axis tracks the subcarrier indices, and the color intensity reflects the CSI magnitude.

Why it's good: It serves as the "fingerprint" of the environment. When a human cuts through the wireless beam, you will see immediate, distinct vertical color shifts or rippling patterns across all subcarriers. It allows you to instantly validate whether an activity or movement was successfully captured in the raw data.



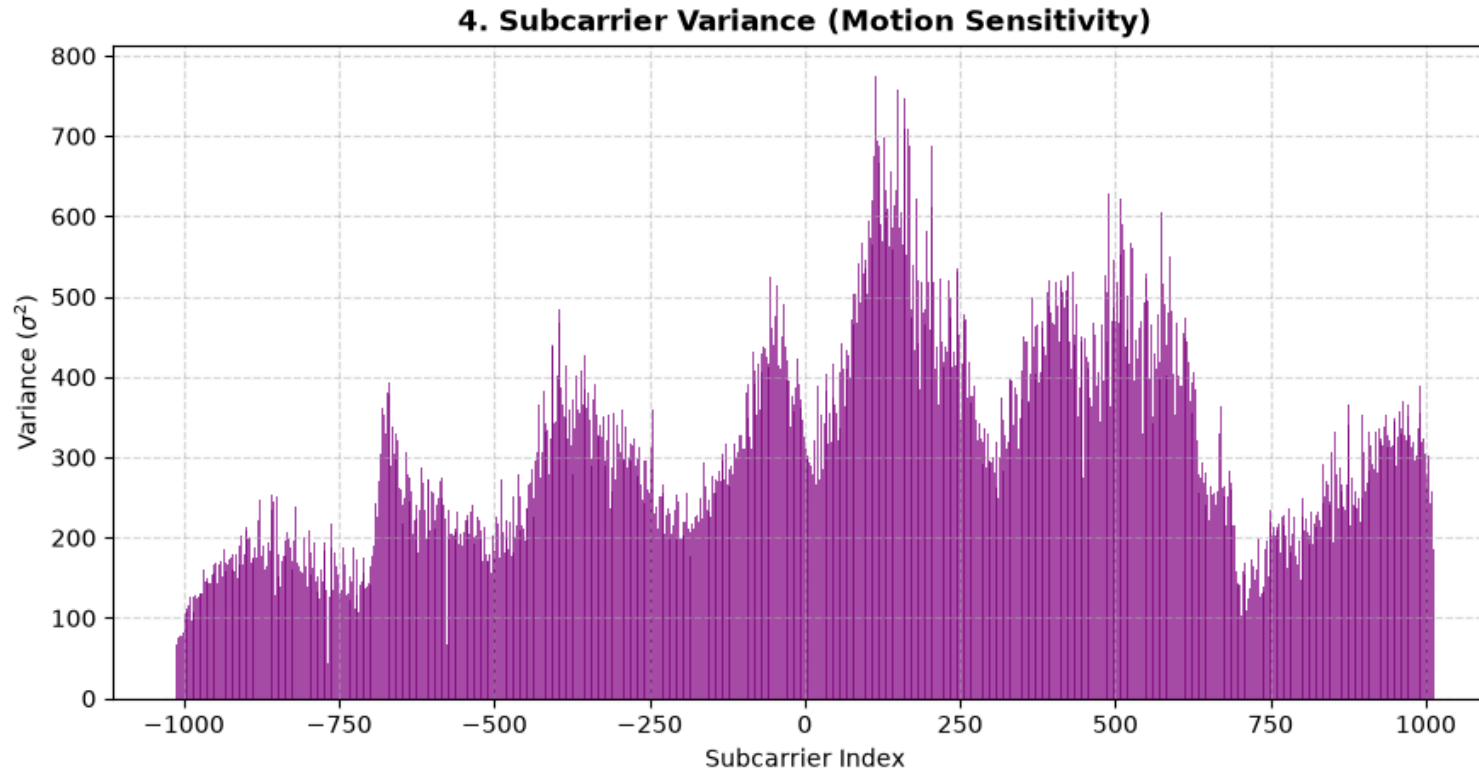
2. Time-Series of Selected Subcarriers

- **What it is:** A microscopic look at a few individual subcarriers isolated across the radio spectrum, tracking their exact magnitude fluctuations over time.
- **Why it's good:** Heatmaps can compress details, but this line graph exposes raw, high-resolution temporal patterns. It is crucial for detecting cyclic, micro-movements. For example, if your subject is sitting still and breathing, this plot will reveal beautiful, rhythmic sinusoidal waves on the most sensitive subcarriers.



3. Frequency Domain Statistical Profile

- **What it is:** A spectral view showing the average Channel Frequency Response (CFR) across the subcarriers, wrapped in a shaded standard deviation band.
- **Why it's good:** Static environments have deep "nulls" (dips in the blue line) due to multipath fading where signals cancel each other out. The thickness of the shaded band tells you how stable those subcarriers are. A wide band means high environmental disruption at those specific frequencies; a narrow band means a stable, unbothered channel.



4. Subcarrier Variance (Motion Sensitivity)

- **What it is:** A bar chart calculating the total statistical variance for every single subcarrier across the entire timeline.
- **Why it's good:** In WiFi-sensing, some subcarriers are highly sensitive to human movement (tall bars), while others are completely blind to it (short/flat bars). This graph tells your Machine Learning pipeline exactly which subcarrier channels to keep for training and which "dead" channels to discard to save computational power.